

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
Data Structure Practical

Practical – II

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Class: SYBSc IT	Subject: DATA STRUCTURE
Date of Assignment: 10/07/2026	Date/Time of Submission: 10/07/2026

2. Linked List Manipulation:

Write a program to:

a. Create a singly linked list.

CODE:

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

node1 = Node(11)
node2 = Node(12)
node3 = Node(14)
node4 = Node(15)

node1.next = node2
node2.next = node3
node3.next = node4

def traver(head):
    curr = head

    while (curr):
        print(curr.data, end="=>")
        curr = curr.next
    print("NULL")
    print("IQRA S021")

traver(node1)
```

OUTPUT:

```
11=>12=>14=>15=>NULL
IQRA S021
```

b. Insert a node at the beginning, end, and at a given position in a linked list.

CODE:

AT BEGINING

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

node1 = Node(11)
node2 = Node(12)
node3 = Node(14)
node4 = Node(15)

node1.next = node2
node2.next = node3
node3.next = node4

global head
head = node1
def traver(head):
    curr = head

    while (curr):
        print(curr.data, end="=>")
        curr = curr.next
    print("NULL")

def insert_beg(data):
    global head
    new = Node(data)
    new.next = node1
    head = new

insert_beg(10)
traver(head)
print("IQRA S021")
```

AT END:

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

node1 = Node(11)
node2 = Node(12)
node3 = Node(14)
node4 = Node(15)

node1.next = node2
node2.next = node3
node3.next = node4

global head
head = node1
def traver(head):
    curr = head

    while (curr):
        print(curr.data, end="=>")
        curr = curr.next
    print("NULL")

def insert_beg(data):
    global head
    new = Node(data)
    new.next = node1
    head = new

def insert_end(data):
    global head
    new = Node(data)

    if head is None:
        head=new
    else :
        temp = head
        while temp.next:
            temp = temp.next
        temp.next = new
insert_beg(10)
insert_end(8)
traver(head)
print("IQRA S021")
```

AT A GIVEN POSITION:

```

temp = head
while temp.next:
    temp = temp.next
temp.next = new

def insert_pos(pos,data):
    global head
    new = Node(data)

    temp = head
    for i in range(pos-2):
        temp = temp.next
    new.next = temp.next
    temp.next = new

insert_beg(10)
insert_end(8)
insert_pos(2,18)
traver(head)
print("IQRA S021")

```

OUTPUT:

```

===== RESIARI: D:/DATA 2
10=>11=>12=>14=>15=>NULL
IQRA S021
|

```

AT END:

```

===== RESIARI: D:/DATA 2
10=>11=>12=>14=>15=>8=>NULL
IQRA S021
|

```

AT A GIVEN POSITION:

```

===== RESIARI: D:/DATA 2
10=>18=>11=>12=>14=>15=>8=>NULL
IQRA S021
> |

```

c. Delete a node from a given position in a linked list.

CODE:

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

node1 = Node(11)
node2 = Node(12)
node3 = Node(14)
node4 = Node(15)

node1.next = node2
node2.next = node3
node3.next = node4

global head
head = node1
def traver(head):
    curr = head

    while (curr):
        print(curr.data, end="=>")
        curr = curr.next
    print("NULL")

def insert_beg(data):
    global head
    new = Node(data)
    new.next = node1
    head = new

def insert_end(data):
    global head
    new = Node(data)

    if head is None:
        head=new
    else :
        temp = head
        while temp.next:
            temp = temp.next
        temp.next = new

def insert_pos(pos, data):
    global head
```

```

print("NULL")

def insert_beg(data):
    global head
    new = Node(data)
    new.next = head
    head = new

def insert_end(data):
    global head
    new = Node(data)

    if head is None:
        head=new
    else :
        temp = head
        while temp.next:
            temp = temp.next
        temp.next = new

def insert_pos(pos,data):
    global head
    new = Node(data)

    temp = head
    for i in range(pos-2):
        temp = temp.next
    new.next = temp.next
    temp.next = new

def delete_pos(pos):
    global head

    temp = head
    for i in range(pos-2):
        temp = temp.next
    temp.next = temp.next.next

insert_beg(10)
insert_end(8)
insert_pos(2,18)
delete_pos(4)
traver(head)
print("IQRA S021")

```

OUTPUT:

```

===== RESTART: D:/
10=>18=>11=>14=>15=>8=>NULL
IQRA S021

```

